

SEQUENCES, SERIES AND DIFFERENTIAL EQUATIONS

Gilberto Berardinelli

Department of Electronic Systems, Aalborg University, Denmark



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Today's agenda

- Derivatives of a function
- Higher order and partial derivatives
- Differential equations
 - Classification
 - First order linear differential equations
 - Separable equations

Derivative of a function

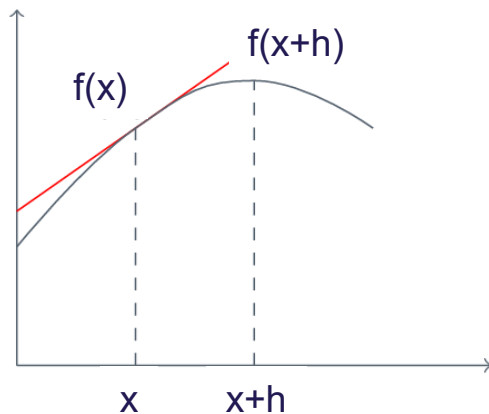


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The **derivative** of a function f is another function f' defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

at all points x for which the limit exists (i.e., is a finite real number). If $f'(x)$ exists, we say that f is **differentiable** at x .



The equation of a tangent line at point x_0 is given by

$$y = f(x_0) + f'(x_0)(x - x_0).$$

Differentiation rules

The Product Rule

If functions f and g are differentiable at x , then their product fg is also differentiable at x , and

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

The Quotient Rule

If f and g are differentiable at x , and if $g(x) \neq 0$, then the quotient f/g is differentiable at x and

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}.$$

Higher order derivatives



$$y'' = f''(x) = \frac{d^2 y}{dx^2} = \frac{d}{dx} \frac{d}{dx} f(x) = \frac{d^2}{dx^2} f(x) = D_x^2 y = D_x^2 f(x).$$

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n} = \frac{d^n}{dx^n} f(x) = D_x^n y = D_x^n f(x),$$

Partial derivatives



The **first partial derivatives** of the function $f(x, y)$ **with respect to the variables** x and y are the functions $f_1(x, y)$ and $f_2(x, y)$ given by

$$f_1(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h},$$

$$f_2(x, y) = \lim_{k \rightarrow 0} \frac{f(x, y + k) - f(x, y)}{k},$$

provided these limits exist.

Notations for first partial derivatives

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} f(x, y) = f_1(x, y) = D_1 f(x, y)$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} f(x, y) = f_2(x, y) = D_2 f(x, y)$$

Partial derivatives

- Chain rule for derivation of functions of one variable

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x).$$

- Chain rule for derivation of functions of multiple variables:

If z is a function of x and y with continuous first partial derivatives, and if x and y are differentiable functions of t , then

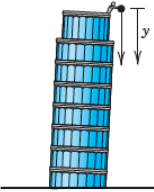
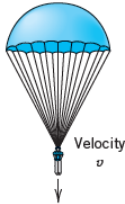
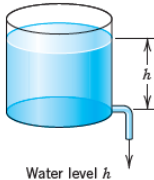
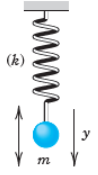
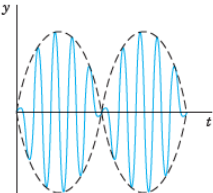
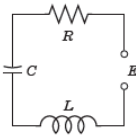
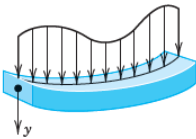
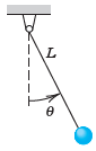
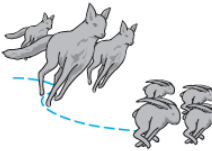
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

Differential equations

Applications from real-world problems



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 Falling stone $y'' = g = \text{const.}$ (Sec. 1.1)	 Parachutist $mv' = mg - bv^2$ (Sec. 1.2)	 Water level h Outflowing water $h' = -k\sqrt{h}$ (Sec. 1.3)
 Displacement y Vibrating mass on a spring $my'' + ky = 0$ (Secs. 2.4, 2.8)	 Beats of a vibrating system $y'' + \omega_0^2 y = \cos \omega t, \quad \omega_0 = \omega$ (Sec. 2.8)	 Current I in an RLC circuit $LI'' + RI' + \frac{1}{C}I = E'$ (Sec. 2.9)
 Deformation of a beam $EIy^{iv} = f(x)$ (Sec. 3.3)	 Pendulum $L\theta'' + g \sin \theta = 0$ (Sec. 4.5)	 Lotka-Volterra predator-prey model $\begin{aligned} y_1' &= ay_1 - by_1y_2 \\ y_2' &= ky_1y_2 - ly_2 \end{aligned}$ (Sec. 4.5)

Differential equations

- A differential equation (or DE) is an equation that involves one or more derivatives of an unknown function

$$\frac{dx}{dt} = g(t). \quad \rightarrow \quad \begin{aligned} x &= G(t) + c, \\ G'(t) &= g(t) \end{aligned} \quad \begin{array}{l} \swarrow \\ \text{arbitrary constant} \end{array}$$

- Functions represents physical functions, and the derivatives represent their rate of change
- An ordinary differential equation (ODE) is one that involves derivatives with respect to only one variable.
- A partial differential equation (PDE) is one that involves partial derivatives of the unknown function with respect to more than one variable.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Differential equations

- Differential equations are also classified with respect to order

$$\frac{d^2 y}{dx^2} + x^3 y = \sin x \quad \text{has order 2,}$$
$$\frac{d^3 y}{dx^3} + 4x \left(\frac{dy}{dx} \right)^2 = y \frac{d^2 y}{dx^2} + e^y \quad \text{has order 3.}$$

- A n-th order linear ODE takes the form

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \dots \\ + a_2(x)y''(x) + a_1(x)y'(x) + a_0(x)y(x) = f(x).$$

- A linear ODE is said to be homogeneous if all of its terms involve the unknown function y , that is, if $f(x)=0$. If $f(x)$ is not identically zero, the equation is nonhomogeneous

Differential equations

- Linear combination of solutions of a homogeneous DE

If $y = y_1(x)$ and $y = y_2(x)$ are two solutions of the linear, homogeneous DE

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0,$$

then so is the linear combination

$$y = Ay_1(x) + By_2(x)$$

for any values of the constants A and B .

If $y = y_1(x)$ is a solution of the linear, homogeneous equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0$$

and $y = y_2(x)$ is a solution of the linear, nonhomogeneous equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x),$$

then $y = y_1(x) + y_2(x)$ is also a solution of the same linear, nonhomogeneous equation.

Solving first order differential equations

- Simplest case

$$\frac{dy}{dx} = f(x). \quad \longrightarrow \quad y = \int f(x) dx.$$

- Separable equations

$$\frac{dy}{dx} = f(x)g(y),$$

It can be solved by rewriting it in the form $\frac{dy}{g(y)} = f(x) dx$, and integrating both sides.

Solving first order differential equations



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EXAMPLE 1 Solve the equation $\frac{dy}{dx} = \frac{x}{y}$.

Solution We rewrite the equation in the form $y \, dy = x \, dx$ and integrate both sides to get

$$\frac{1}{2} y^2 = \frac{1}{2} x^2 + C_1,$$

or $y^2 - x^2 = C$, where $C = 2C_1$ is an arbitrary constant. The solution curves are rectangular hyperbolas. (See Figure 7.64.) Their asymptotes $y = x$ and $y = -x$ are also solutions corresponding to $C = 0$.

